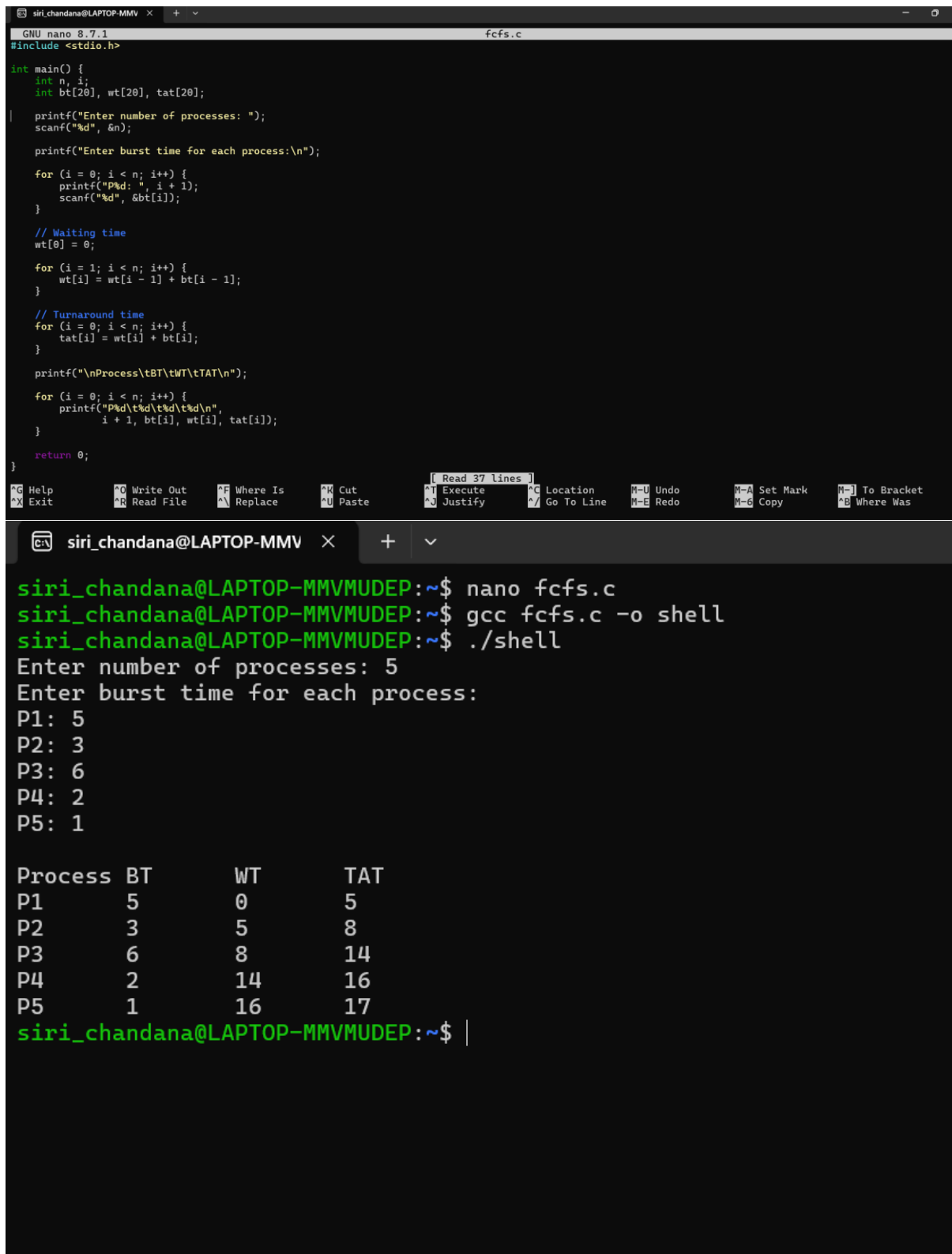


1.



```
GNU nano 8.7.1 fcfs.c
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Waiting time
    wt[0] = 0;
    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

siri\_chandana@LAPTOP-MMV × + ▾

siri\_chandana@LAPTOP-MMV:~\$ nano fcfs.c  
siri\_chandana@LAPTOP-MMV:~\$ gcc fcfs.c -o shell  
siri\_chandana@LAPTOP-MMV:~\$ ./shell

Enter number of processes: 5  
Enter burst time for each process:  
P1: 5  
P2: 3  
P3: 6  
P4: 2  
P5: 1

| Process | BT | WT | TAT |
|---------|----|----|-----|
| P1      | 5  | 0  | 5   |
| P2      | 3  | 5  | 8   |
| P3      | 6  | 8  | 14  |
| P4      | 2  | 14 | 16  |
| P5      | 1  | 16 | 17  |

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2.

GNU nano 8.7.1

#include &lt;stdio.h&gt;

```
int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
```

^G Help  
^X Exit

^O Write Out  
^R Read File

^F Where Is  
^\_ Replace

^K Cut  
^U Paste



GNU nano 8.7.1

```
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
               i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
siri_chandana@LAPTOP-MMVMUDEP:~$ nano sjf.c
siri_chandana@LAPTOP-MMVMUDEP:~$ gcc sjf.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~$ ./shell
Enter number of processes: 3
Enter burst time:
P1: 3
P2: 6
P3: 2

Process BT      WT      TAT
P1      2      0      2
P2      3      2      5
P3      6      5     11
siri_chandana@LAPTOP-MMVMUDEP:~$ |
```

3.

```
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    // Sort according to priority
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {

                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
```

GNU nano 8.7.1

```
for (i = 0; i < n - 1; i++) {
    for (j = i + 1; j < n; j++) {
        if (priority[i] > priority[j]) {

            temp = priority[i];
            priority[i] = priority[j];
            priority[j] = temp;

            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
        }
    }
}

// Waiting time
wt[0] = 0;

for (i = 1; i < n; i++) {
    wt[i] = wt[i - 1] + bt[i - 1];
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t\t%d\t%d\n",
        i + 1, bt[i], priority[i], wt[i], tat[i]);
}

return 0;
}
```

```
siri_chandana@LAPTOP-MMV  ×  +  ∨  
siri_chandana@LAPTOP-MMVUDEP:~$ nano priority.c  
siri_chandana@LAPTOP-MMVUDEP:~$ gcc priority.c -o shell  
siri_chandana@LAPTOP-MMVUDEP:~$ ./shell  
Enter number of processes: 3  
Enter burst time and priority:  
P1 Burst Time: 11  
P1 Priority: 11  
P2 Burst Time: 12  
P2 Priority: 1  
P3 Burst Time: 5  
P3 Priority: 18  
  
Process BT      Priority      WT      TAT  
P1      12      1      0      12  
P2      11      11      12      23  
P3      5      18      23      28  
siri_chandana@LAPTOP-MMVUDEP:~$ |
```

4.

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 roundrobin.c *
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    // Round Robin
    while (1) {
        done = 1;

        for (i = 0; i < n; i++) {
            if (rem_bt[i] > 0) {
                done = 0;

                if (rem_bt[i] > quantum) {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
            }
        }
    }
}
```

**Help**   **Write Out**   **Where Is**   **Cut**   **Execute**   **Loca**  
**Exit**   **Read File**   **Replace**   **Paste**   **Justify**   **Go**

```
siri_chandana@LAPTOP-MMV  ×  +  v
GNU nano 8.7.1 roundrobin.c
while (1) {
    done = 1;

    for (i = 0; i < n; i++) {
        if (rem_bt[i] > 0) {
            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            } else {
                time = time + rem_bt[i];
                wt[i] = time - bt[i];
                rem_bt[i] = 0;
            }
        }
    }

    if (done == 1)
        break;

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
```

```
siri_chandana@LAPTOP-MMVMUDEP:~$ nano roundrobin.c
siri_chandana@LAPTOP-MMVMUDEP:~$ gcc roundrobin.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~$ ./shell
Enter number of processes: 2
Enter burst time:
P1: 12
P2: 6
Enter time quantum: 2

Process BT      WT      TAT
P1      12      6       18
P2       6      6       12
siri_chandana@LAPTOP-MMVMUDEP:~$ |
```